

$\frac{5}{5}$

Q1) Determine whether or not each of the following signals is periodic and specify its fundamental period?

1-

~~XXXXXX~~ $\frac{5}{5}$

$$2) T_1 = \frac{2\pi}{10} = \frac{\pi}{5}$$

$$T_2 = \frac{2\pi}{4} = \frac{\pi}{2}$$

$$\left\{ \begin{array}{l} T_1 = \frac{\pi}{5}, \frac{2\pi}{5}, \frac{4\pi}{5}, \dots \\ T_2 = \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \dots \end{array} \right. \left[\frac{2\pi}{5} \right]$$

$$\frac{T_1}{T_2} = \frac{2}{5} \quad \text{periodic} \checkmark$$

OR

2-

$$T_{\text{new period}} = 2\pi$$

$\frac{2\pi}{5}$

$$\frac{2\pi}{5} \cdot 5 = 2\pi$$

$$x(t) = 2\cos(10t+1) - \sin(4t-1)$$

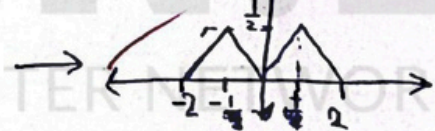
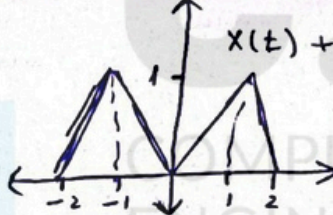
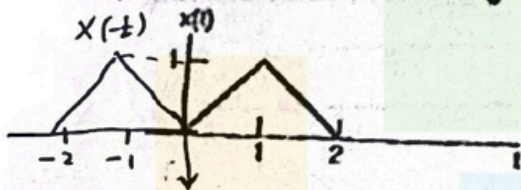
Q2) Determine and sketch the even and odd parts of the signals?

1-

$$X_e(t) = \frac{1}{2} [X(t) + X(-t)]$$

$$X(t) + X(-t)$$

$$\frac{1}{2} [X(t) + X(-t)]$$



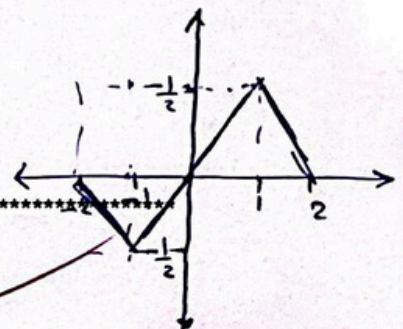
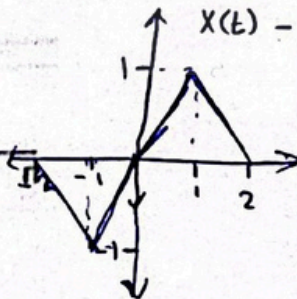
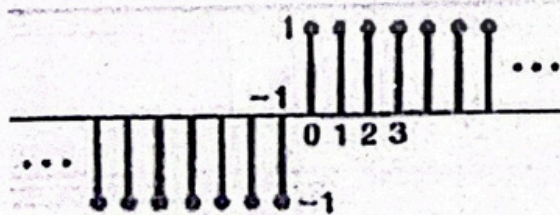
OR

2-

$$X_o(t) = \frac{1}{2} [X(t) - X(-t)]$$

$$X(t) - X(-t)$$

$$\frac{1}{2} [X(t) - X(-t)]$$



$$\cos(t+nT) = \cos(t+nT)$$

$$T = \frac{2\pi}{1}$$

$$T = 2\pi$$

$$= \cos t \cos nT + \sin t \sin nT$$

$$= \cos t \cos(n \times 2\pi) + \sin t \sin(n \times 2\pi)$$

$$\cos(2\pi n) = \cos t \text{ periodic}$$

Q3) Determine if the following signals are power or Energy?

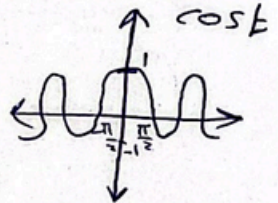
1-

$$x_1(t) = e^{-2t} u(t)$$

$$= \begin{cases} e^{-2t}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$2) x_3(t) = \cos t$$

$\cos t \rightarrow$ periodic signal
So it's a power energy



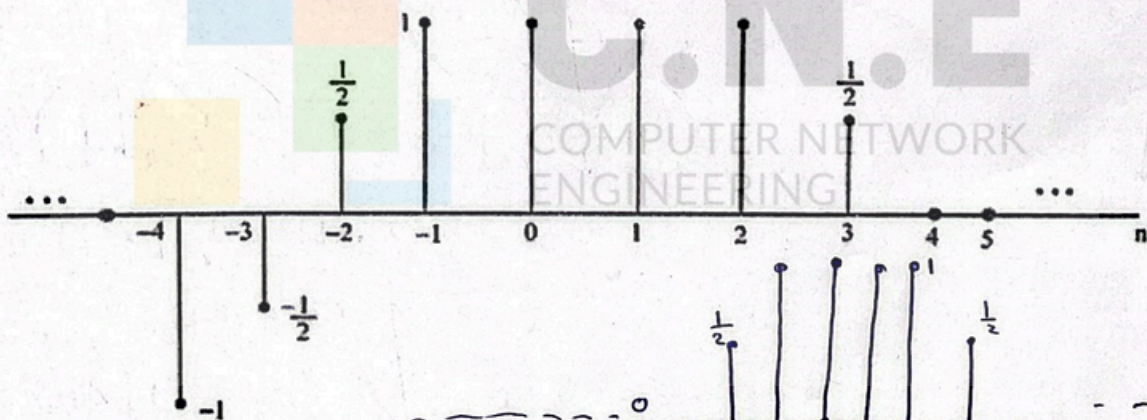
OR

2-

$$x_3(t) = \cos t$$

Q4)

Consider the following signal $x[n]$:



Find

$$x[n-4]$$

Q5) Draw the following signal:

$$f(t) = u(t+2) - u(t-1)$$

